

Figure5A

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2025-08-06

Intro

Compares the NTI values between semesters to reproduce the results of Figure 5A from the original publication Seasonal dynamics of the coastal microbiome and its association with environmental factors.

1. Set the environment

```
library(tidyverse)
library(vegan)
```

2. Load data

NTI_weighted_26_1000_asvs.csv weighted Nearest Taxin Index (NTI) values for the 1000 most abundant ASVs in the dataset. samo_metadata_workable.tsv contains the metadata for the samples. date2season2community.tsv is a table mapping the date, season, and community columns.

```
NTI <- read_csv("../data/tree_analysis/NTI_weighted_26_1000_asvs.csv", show_col_types = FALSE)
```

```
## New names:
## * `` -> `...1`
```

```
colnames(NTI)[1] <- "sample_name"
METADATA <- read_tsv("../data/samo_metadata_workable.tsv", show_col_types = FALSE)
DATE2SEASON2COMMUNITY <- read_tsv("../data/date2season2community.tsv", show_col_types = FALSE)
```

3. Format and add metadata info to the NTI table

```
NTI_ext <- NTI %>%
  filter(sample_name %in% METADATA$sample_name) %>%
  left_join(x = ., y = METADATA, by = "sample_name") %>%
  left_join(x = ., y = DATE2SEASON2COMMUNITY, by = "Date") %>%
  mutate(mntd.obs.z = -mntd.obs.z) %>%
  mutate(bnti_signif = if_else(mntd.obs.z > 2, "gt",
                               if_else(mntd.obs.z < -2, "lt", "not_signif")))
```

4. Get percentage of samples with nti greater than 2

```
NTI_percs <- NTI_ext %>%
  group_by(Community) %>%
  summarise(n = n(),
            n_bnti_gt_2 = sum(mntd.obs.z > 2),
            n_bnti_lt_2 = sum(mntd.obs.z < -2),
            perc_bnti_gt_2 = n_bnti_gt_2/n,
```

```

      perc_bnti_lt_2 = n_bnti_lt_2/n) %>%
    arrange(desc(n))

nti_perc_s1 <- NTI_percs$perc_bnti_gt_2[NTI_percs$Community == "S1"]
nti_perc_s2 <- NTI_percs$perc_bnti_gt_2[NTI_percs$Community == "S2"]

```

5. Create NTI boxplot

```

scaling_factor <- 15
scaling_constat <- -50
text_size <- 12

nti_boxplots <- ggplot(NTI_ext, aes(x = Community,
                                   y = mntd.obs.z)) +
  xlab("Semester") +
  ylab("NTI") +
  scale_fill_manual(values = c("#b14b34", "#292c33")) +
  geom_hline(yintercept = 2, linetype = "dashed") +
  geom_hline(yintercept = -2, linetype = "dashed") +
  scale_x_discrete(labels = c("S1" = "Summer-Autumn",
                              "S2" = "Winter-Spring")) +
  theme_bw() +
  theme(
    axis.text.x = element_text(size = text_size, margin = margin(b = 5)),
    axis.text.y = element_text(size = text_size),
    axis.title = element_text(size = text_size + 4),
    strip.text = element_text(size = text_size + 4),
    strip.background = element_blank(),
  ) +
  guides(fill = "none") +
  geom_boxplot(alpha = 0.8, aes(fill = Community)) +
  geom_text(data = NTI_percs, aes(x = Community, y = 3.5,
                                  label = scales::percent(perc_bnti_gt_2, accuracy = 0.01),
                                  vjust = -0.5, size = 2.5))

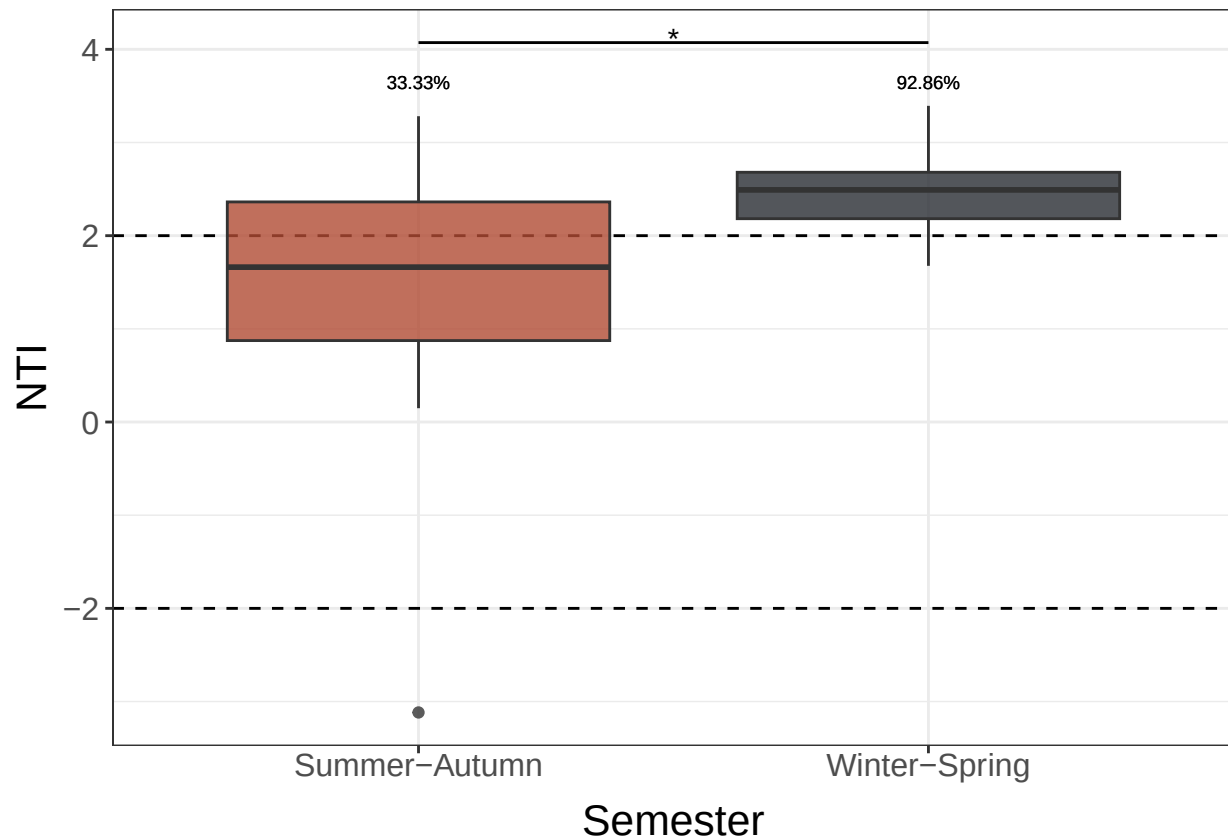
y_pos <- max(NTI_ext$mntd.obs.z, na.rm = TRUE) * 1.2

nti_boxplots_ext <- nti_boxplots +
  geom_text(data = NTI_percs,
            aes(x = Community, y = 3.5,
                label = scales::percent(perc_bnti_gt_2, accuracy = 0.01)),
            vjust = -0.5, size = 2.5) +
  # ---- significance bar ----
  annotate("segment",
    x = 1, xend = 2,
    y = y_pos, yend = y_pos,
    size = 0.5) +
  annotate("text",
    x = 1.5, y = y_pos,
    label = "*",
    vjust = 0.3,
    size = 4)

```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was generated.
```

```
nti_boxplots_ext
```



6. Print session info

```
sessionInfo()
```

```
## R version 4.4.2 (2024-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 20.04.6 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.9.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.9.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C              LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
##  [6] LC_MESSAGES=en_US.UTF-8  LC_PAPER=en_US.UTF-8     LC_NAME=C                LC_ADDRESS=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Etc/UTC
## tzcode source: system (glibc)
##
## attached base packages:
```

```
## [1] grid      stats      graphics  grDevices utils      datasets  methods  base
##
## other attached packages:
## [1] fANCOVA_0.6-1      maps_3.4.2          corrgram_1.14       ggpubr_0.6.0        ggcorrplot_0.1.4.1
## [8] permute_0.9-7      lubridate_1.9.3     forcats_1.0.0       stringr_1.5.1       dplyr_1.1.4
## [15] tidyr_1.3.1        tibble_3.2.1        ggplot2_3.5.1       tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] gtable_0.3.5      xfun_0.48           rstatix_0.7.2       tzdb_0.4.0          vctrs_0.6.5         tools
## [9] fansi_1.0.6       highr_0.11          cluster_2.1.6        pkgconfig_2.0.3     Matrix_1.7-0        lifecycle
## [17] munsell_0.5.1     carData_3.0-5       htmltools_0.5.8.1   yaml_2.3.10         Formula_1.2-5       crayon
## [25] MASS_7.3-61       abind_1.4-8         nlme_3.1-166        tidyselect_1.2.1    digest_0.6.37       stringr
## [33] cowplot_1.1.3     fastmap_1.2.0       colorspace_2.1-1    cli_3.6.3           magrittr_2.0.3      utf8
## [41] scales_1.3.0      backports_1.5.0     bit64_4.5.2         timechange_0.3.0    rmarkdown_2.28      bit_4
## [49] evaluate_1.0.1    knitr_1.48          mgcv_1.9-1          rlang_1.1.4         glue_1.8.0          vroom
```